Application of the three-Coulomb-wave model with variable charge (3CWZ) to the single ionization of CH_4 and NH_3 by electron impact

Final report of the research protocol — DRAFT

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Abstract

We extend the three-Coulomb-wave model with a position-dependent effective charge (3CWZ) from atomic to molecular targets and apply it to the single ionization of CH_4 and NH_3 by electron impact.

- **Target:** described by the one-centre SCF molecular orbitals of Moccia. The random orientation of the molecule is averaged analytically.
- **Effective charges:** derived from the monopole part of the Hartree potential of the SCF density, with a self-interaction correction for the ejected electron.
- **Code:** we implemented the model in a parallel Fortran code. We validated it against the original CH_4 code and against analytic limits.
- **Comparisons for NH_3:**
  - triple differential cross sections measured by El Mir *et al.* at $E_s = 500$ eV, $\theta_s = -6°$, $E_e = 74$ eV;
  - electron momentum spectroscopy profiles (1200 eV);
  - double differential cross sections at $E_i = 500$ eV.

The binary peak is well reproduced for the three valence orbitals $3a_1$, $1e$ and $2a_1$. The recoil peak is underestimated by a factor 5–10. We trace this deficit to the ejected-electron wave: the variable-charge Coulomb wave does not solve the one-electron equation in the screened ionic potential, and it misses the strong back-scattering by the NH_3^+ core. We delimit the kinematic and numerical domain of validity of the model, and we propose a distorted-wave treatment of the ejected electron as the next step.

To complete: Update the abstract with the CH_4 results, the PCI study and the variable- versus constant-charge comparison (Sections 4.6–4.8).

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1 Introduction

Electron-impact ionization is a basic process in collision physics. It is also a key input for:

- plasma modelling;
- atmospheric and astrophysical chemistry;
- radiation–matter interaction, where low-energy secondary electrons produced by ionizing radiation cause further ionization and fragmentation of biomolecules.

Kinematically complete $(e, 2e)$ experiments measure the triple differential cross section (TDCS). The TDCS is sensitive both to the structure of the target and to the dynamics of the three charged particles in the final state.

Theoretical descriptions form a hierarchy of approximations:

- the plane-wave Born approximation (PWBA);
- distorted-wave methods (DWBA, M3DW);
- three-Coulomb-wave models (3C or BBK) [1], which treat the three two-body Coulomb interactions of the final state on an equal footing.

The 3CWZ model [2, 3, 4] replaces the constant charges of the 3C model by a position-dependent effective charge $Z(r)$. This charge interpolates between the nuclear charge near the nucleus and the ionic charge far from it. It has improved the description of atomic targets.

Molecules add several difficulties:

- the potential is multi-centre and non-spherical;
- the orientation of the molecule is random;
- the orbital structure is more complex.

CH_4 and NH_3 are ten-electron hydrides whose C–H and N–H bonds are building blocks of biological molecules. CH_4 is nearly spherical. NH_3 is more anisotropic and has a permanent dipole moment.

The objectives of the protocol were:

O1. adapt the 3CWZ formalism to molecular targets with the one-centre orbitals of Moccia, including $Z(r)$ for each orbital;

O2. compute TDCS for the orbitals of CH_4 and NH_3 under various kinematics;

O3. study the post-collision interaction (PCI) by comparing results with and without the electron–electron factor C_{ab};

O4. validate the model against the available experimental and theoretical data.

The expected outcomes were to show what the variable charge contributes and to delimit the domain of validity of the model for molecules.

The report is organised as follows:

- Section 2: the formalism;
- Section 3: the numerical implementation and its validation;
- Section 4: the results;
- Section 5: the origin of the recoil-peak deficit and the domain of validity;
- Section 6: conclusions and perspectives.

Atomic units are used throughout unless stated otherwise.

2 Theory

2.1 Kinematics and cross section

We consider $e^-(\mathbf{k}_i) + M \to M^+ + e^-(\mathbf{k}_s) + e^-(\mathbf{k}_e)$, with

$$E_i = E_s + E_e + I_n, \qquad \mathbf{K} = \mathbf{k}_i - \mathbf{k}_s, \qquad \mathbf{q} = \mathbf{K} - \mathbf{k}_e, \tag{1}$$

where:

- I_n is the binding energy of the ionized orbital;
- $\mathbf{K}$ is the momentum transfer;
- $\mathbf{q}$ is the recoil momentum of the ion.

The TDCS is

$$\sigma^{(3)} = \frac{d^3\sigma}{d\Omega_s\, d\Omega_e\, dE_e} = (2\pi)^4 \frac{k_s k_e}{k_i} |T_{fi}|^2. \tag{2}$$

At the energies considered here ($E_s \gg E_e$), exchange is neglected and $T_{fi} = T_{dir}$.

To complete: If the low-energy data of Nixon *et al.* [12] are analysed, include the exchange amplitude. Check the factor $\frac{1}{2}$ of the spin average: $\frac{1}{2}\left(|T_d|^2 + |T_e|^2 + |T_d - T_e|^2\right)$.

2.2 The 3CWZ final state

The final state is the product of three Coulomb waves:

$$\Psi_f = \phi_s^-(\mathbf{k}_s, \mathbf{r}_0)\, \phi_e^-(\mathbf{k}_e, \mathbf{r}_1)\, C_{se}^-(\mathbf{k}_{se}, \mathbf{r}_{01}), \qquad \mathbf{k}_{se} = \tfrac{1}{2}(\mathbf{k}_s - \mathbf{k}_e), \tag{3}$$

Each electron–ion wave has the form

$$\phi_c^{\pm}(\mathbf{k}, \mathbf{r}) = (2\pi)^{-3/2}\, e^{\pi\alpha(r)/2}\, \Gamma(1 \mp i\alpha(r))\, e^{i\mathbf{k}\cdot\mathbf{r}}\, {}_1F_1(\pm i\alpha(r); 1; \pm i(kr \mp \mathbf{k}\cdot\mathbf{r})), \qquad \alpha(r) = \frac{Z(r)}{k}. \tag{4}$$

- The effective charge $Z(r)$ goes to 1 at large r for the scattered and ejected electrons, and to 0 for the incident electron (neutral target).
- The incident wave $\phi_i^+(\mathbf{k}_i, \mathbf{r}_0)$ uses the neutral-target charge.
- The factor C_{se} describes the electron–electron interaction (PCI), with $\alpha_{se} = -1/(2k_{se})$.

In the frozen-core, one-active-electron approximation the interaction is $V = -1/r_0 + 1/r_{01}$, and the direct amplitude is

$$T_{dir} = \left\langle \Psi_f \middle| \frac{1}{r_{01}} - \frac{1}{r_0} \middle| \phi_i^+\, \Phi \right\rangle = T^1 - T^0. \tag{5}$$

- The six-dimensional integrals are made separable by a Fourier representation of C_{se}.
- Each wave is split into a $Z = 1$ (or $Z = 0$) Coulomb part, which is integrated analytically, and a short-range correction confined to $r < d$, which is integrated numerically.
- The cut-off is $d = 4$ a.u. It is justified by $Z_{ion}(4) = 1.011$ (Section 4.1).

The full derivation is given in the 2nd-quarter report.

2.3 Molecular target and orientation average

The molecular orbitals are the one-centre SCF orbitals of Moccia [5, 6]. They are expanded on Slater functions centred on the heavy atom:

$$\Phi(\mathbf{r}) = \sum_j c_j \, \frac{(2\zeta_j)^{n_j+1/2}}{\sqrt{(2n_j)!}} \, r^{n_j-1} e^{-\zeta_j r} \, S_{l_j m_j}(\hat{\mathbf{r}}), \tag{6}$$

where S_{lm} are real spherical harmonics. For NH_3 we use Table I of Moccia II: pyramidal geometry, $R_{\mathrm{N-H}} = 1.928$ a.u., $\theta = 108.9°$, with the valence orbitals $2a_1$, $1e$ and $3a_1$ (vertical ionization energies 27.0, 16.4 and 10.85 eV).

The amplitude is linear in the orbital. Each orbital is therefore split into parts g of given (l_g, m_g), $\Phi = \sum_g R_g(r) S_{l_g m_g}$. The orthogonality of the rotation matrices,

$$\frac{1}{8\pi^2} \int d\Omega_{\mathrm{Euler}} \, D^{l}_{\mu m} D^{l'*}_{\mu' m'} = \frac{\delta_{ll'}\delta_{mm'}\delta_{\mu\mu'}}{2l+1}, \tag{7}$$

then gives the orientation-averaged TDCS exactly:

$$\langle \sigma^{(3)} \rangle = (2\pi)^4 \frac{k_s k_e}{k_i} \sum_g \frac{1}{2l_g+1} \sum_{\mu=-l_g}^{l_g} \left| T[R_g Y_{l_g \mu}] \right|^2. \tag{8}$$

2.4 Variable effective charge

Following the 3rd-quarter report, $Z(r) = r\,V_{\mathrm{total}}(r)$, where V_{total} is the monopole ($L = 0$) part of the electrostatic potential of the target. It has three contributions:

- the heavy nucleus;
- the Hartree potential of the SCF electron density;
- the hydrogen nuclei, modelled as a uniform spherical shell of charge n_H at radius a_H.

For each pair of Slater functions of the same (l, m) part, the screening integral has the closed form

$$b(r) = \frac{p!}{c^{p+1}} - J_p(c, r) + r\, J_{p-1}(c, r), \qquad J_n(\alpha, r) = \int_r^\infty t^n e^{-\alpha t} dt, \tag{9}$$

with $p = n_i + n_j$ and $c = \zeta_i + \zeta_j$. Then

$$Z(r) = Z_N - \sum_{\mathrm{MO}} n_{\mathrm{MO}} \sum_{\mathrm{pairs}} c_i c_j N_i N_j \, b(r) + n_H \min(r/a_H, 1). \tag{10}$$

- **Incident electron:** sees the full density, so $Z \to Z_N - N_e + n_H = 0$.
- **Ejected and scattered electrons:** one electron is removed from the ionized orbital (self-interaction correction), so $Z \to 1$.

3 Numerical implementation and validation

3.1 Code

The code `nh3_opt_claude.for` generalizes the CH_4 3CWZ code to any XH_n target described by Slater-type one-centre orbitals. Its main features are:

- The target data (nuclear charge, proton shell, all occupied orbitals and their (l, m) parts) are given in one routine. The ionized orbital is selected by a switch.

- $Z(r)$ is computed from these data for the neutral molecule and for the ion.
- The $Z = 1$ parts of the ejected-electron integrals are analytic. The short-range corrections ($r < d$) use Gauss–Legendre quadrature: 4 radial intervals $\times$ 10 points, and 10×10 angular points.
- The Fourier integral over $\mathbf{p}$ uses $60 \times 40 \times 40$ points, refined around $|\mathbf{p}| = K$.
- In coplanar geometry, the mirror symmetry $y \to -y$ halves the cost.
- The code is parallelized with OpenMP. A coplanar angular distribution (61 ejection angles) takes about 3 min on 4 cores.

3.2 Validation

- With the CH_4 target, the code reproduces the original CH_4 code exactly.
- Analytic and numerical evaluations of the $Z = 1$ ejected-electron integral agree to 10^{-7}.
- The mirror-symmetric and full angular grids give identical results.
- All molecular orbitals are normalised to 1 ± 10^{-5}.
- $Z(r)$ reaches its asymptotic values 0 (neutral) and 1 (ion).

3.3 Options tested

- `isr=1`: the projectile–core interaction $-1/r_0$ is replaced by the screened static potential $-Z_{ion}(r_0)/r_0$, i.e. its short-range part is added.
- `iorth=1`: the ejected wave is Schmidt-orthogonalized to the ionized orbital, partial wave by partial wave, before the amplitude is computed.

3.4 Numerical domain of validity

We repeated calculations with finer grids: 16 points per interval, 8 radial intervals for the projectile and 4 for the ejected electron. The default grids are accurate to about 20% when both outgoing electrons are slow and the momentum transfer is small ($K \lesssim 3$ a.u.). This holds, for example, for the El Mir kinematics ($K \approx 0.85$ a.u., $E_e = 74$ eV). Outside this domain they are not accurate:

- for a 600 eV ejected electron (EMS kinematics), the TDCS changes by up to +70%;
- at large momentum transfer ($\theta_s \gtrsim 40°$ at $E_i = 500$ eV), the TDCS is overestimated by factors up to 30.

Integrated cross sections, which sample large K, therefore need grids adapted to K.

To complete: Implement K-adapted grids, or switch to the analytic $Z = 1$ limit beyond $K \approx 3$ a.u., and repeat the convergence study.

4 Results

4.1 Effective charges

Figure 1 and Table 1 show the computed charges.

- Both curves start from Z_N: 6 for CH_4, 7 for NH_3.
- They fall steeply inside $r \approx 1$ a.u.

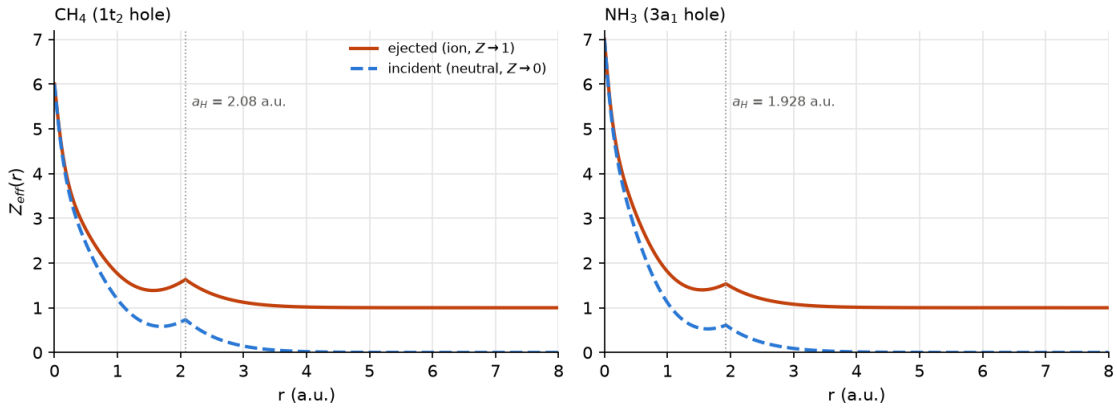


Figure 1: Effective charge $Z(r)$ of CH_4 ($1t_2$ hole) and NH_3 ($3a_1$ hole) for the ejected electron (ion) and the incident electron (neutral molecule). The dotted line marks the X–H distance a_H.

- They show a local maximum at the X–H distance, where the proton shell stops contributing.
- They reach their asymptotes by $r \approx 4$ a.u.
- The ejected-electron charge lies above the incident one at every r; this is the self-interaction correction.
- The three NH_3 orbitals give nearly identical $Z_{ion}(r)$ (within 0.05).

Table 1: $Z(r)$ for NH_3 from the code.

r (a.u.)	0	0.5	1.0	1.5	2.0	2.5	3.0	4.0
incident (neutral)	7.000	2.690	1.113	0.549	0.541	0.227	0.091	0.013
ejected ($3a_1$ hole)	7.000	3.094	1.806	1.402	1.474	1.198	1.079	1.011

4.2 NH_3: TDCS in asymmetric coplanar kinematics

We consider the conditions of El Mir *et al.* [7]: $E_s = 500$ eV, $\theta_s = -6°$, and $E_e = 74$ eV. The incident energy follows from $E_i = E_s + E_e + I_n$. The experimental data were digitized from the published figures. Being on a relative scale, each calculation is scaled to the maximum of the experimental binary peak. The momentum transfer points at $\theta_K \approx 46$–$50°$.

Table 2: Peak positions and binary-to-recoil ratios at $E_e = 74$ eV.

Orbital		binary peak	recoil peak	binary/recoil
$3a_1$	experiment	54°	242°	2.1
	3CWZ	60°	flat	14.8
1e	experiment	49°	240°	2.5
	3CWZ	54°	flat	23.4
$2a_1$	experiment	50°	230°	1.1
	3CWZ	48°	246° (weak)	5.3

Figure 2 and Table 2 show the following:

- The position and the front side of the binary peak are reproduced for all three orbitals.
- The recoil region is underestimated by a factor 5–10. So is the large-angle side of the binary peak: at $\theta_e = 120°$, the 1e and $3a_1$ data are about 7 times above the model.

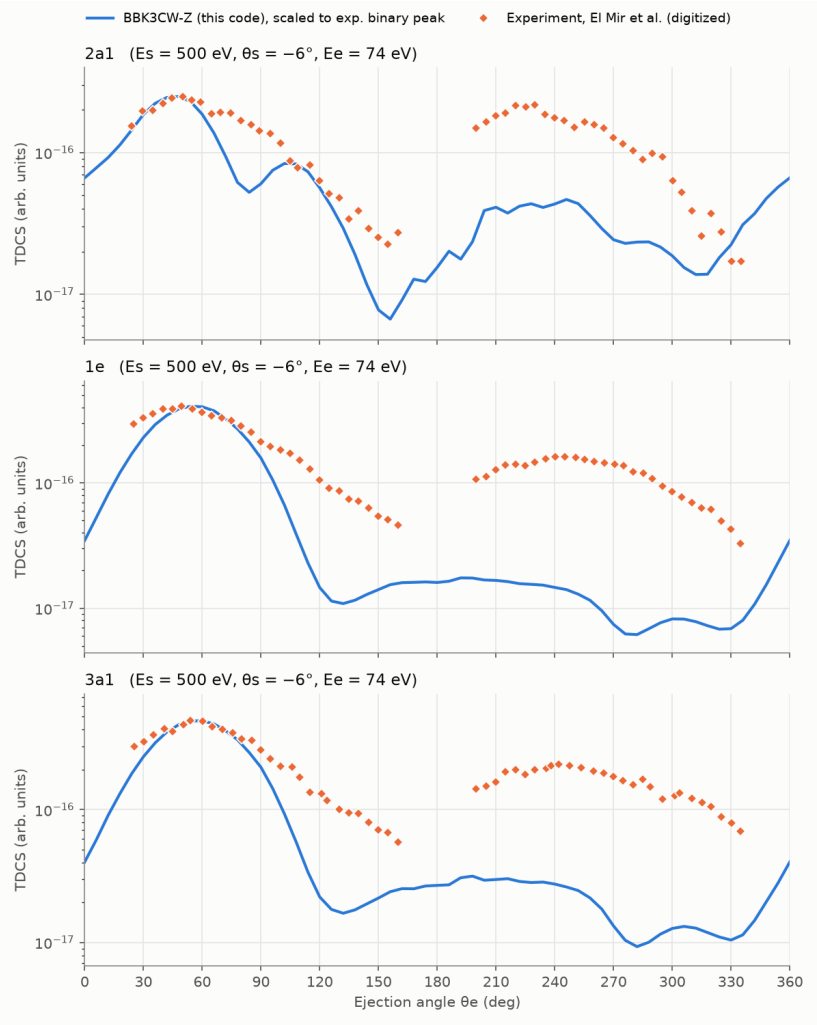


Figure 2: TDCS for the $2a_1$, 1e and $3a_1$ orbitals of NH_3 at $E_s = 500$ eV, $\theta_s = -6°$, $E_e = 74$ eV: 3CWZ (line) and experiment [7] (symbols).

- For $2a_1$ the model shows a dip at about 80°, the node of the 2s-like momentum distribution, which the experiment does not show.

To complete: Digitize the $E_e = 37$ eV data of [7] and compare with the calculations already done at that energy.

4.3 Short-range projectile potential and orthogonalization

Short-range potential (`isr=1`, Fig. 3, left).

- It raises the cross section at large angles, giving binary/recoil ratios of 0.98, 1.8 and 0.7 for $3a_1$, 1e and $2a_1$.
- The rise is nearly isotropic, a shelf rather than a peak centred on $-\mathbf{K}$.
- It shifts the binary peak forward to 60–66° and distorts it for $2a_1$.

The ejected Coulomb wave is not orthogonal to the bound orbital. The overlaps are 0.37 for the s part of $2a_1$, and 0.10 (s) and 0.18–0.20 (p) for $3a_1$. Through these overlaps the projectile–core term produces this spurious isotropic contribution.

Orthogonalization (`iorth=1`, Fig. 3, right).



Figure 3: Left: 3CWZ without (dashed) and with (solid) the short-range projectile–core potential. Right: 3CWZ, orthogonalized ejected wave, and orthogonalized wave with the short-range potential. Same kinematics as Fig. 2.

- It removes the shelf, and the $2a_1$ binary peak recovers its shape.
- The recoil region drops back to about 10 times below the data.

The physical back-scattering contained in the 3CWZ ejected wave is therefore too weak.

4.4 NH_3: electron momentum spectroscopy

As a test of the orbitals and of the high-energy limit, we computed the momentum profiles measured by Bawagan *et al.* [8]: $E_0 = 1200$ eV+I_n, symmetric non-coplanar geometry, $\theta = 45°$, 600 eV each (Fig. 4).

- The peak positions are reproduced (q ≈ 0.58, 0.1 and 0.57 a.u.), and so are the shapes for $2a_1$ and $3a_1$.
- The 1e profile is about 1.5 times low above $q \approx 1.3$ a.u. The paper's PWBA shows the same, so the cause is the high-momentum content of the 1e orbital.
- The absolute values with the default grids are 0.57 of the PWBA of Rezkallah *et al.* [10]. With finer grids, $3a_1$ reaches 0.76; see Section 3.4.

4.5 NH_3: double differential cross section

We integrated the TDCS over the scattered-electron direction:

- 17 points equally spaced in $\ln K$ and 5 azimuths;
- the three valence orbitals summed, with 1e counted twice.

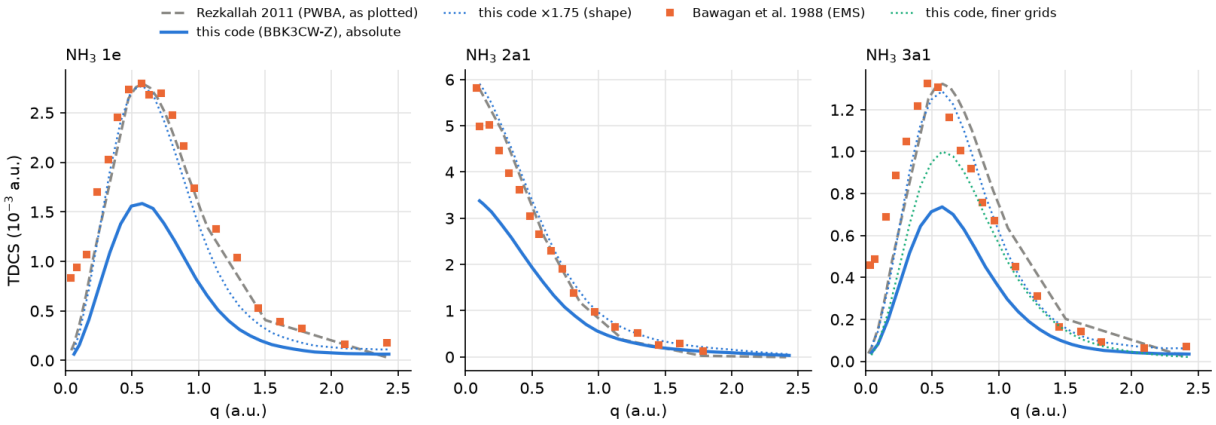


Figure 4: NH_3 momentum profiles at 1200 eV: 3CWZ (absolute, and scaled by 1.75), PWBA of Ref. [10], and experiment [8].

This gives the DDCS at $E_i = 500$ eV and $\theta_e = 120°$ (Fig. 5). The model exceeds the data of Opal *et al.* [9] by factors of 1.79, 1.58 and 1.35 at $E_e = 5$, 20 and 60 eV. At 150 eV the excess (factor 7) comes from large momentum transfers, where the default grids are not converged (Section 3.4).

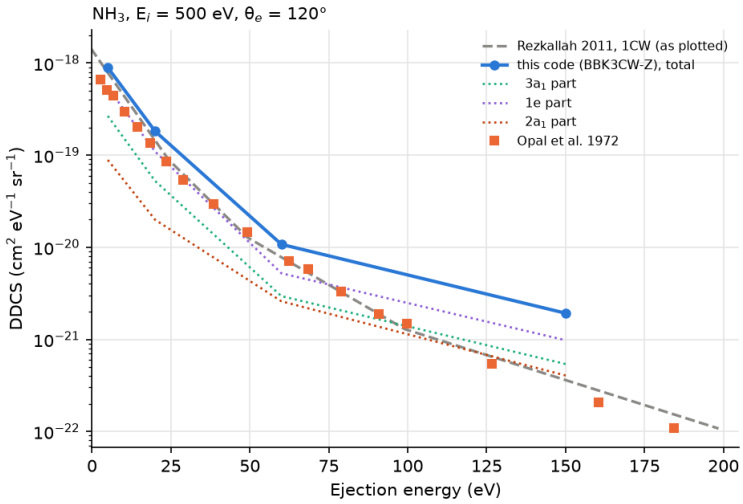


Figure 5: DDCS of NH_3 at $E_i = 500$ eV, $\theta_e = 120°$: 3CWZ (total and per orbital), 1CW of Ref. [10], and experiment [9].

4.6 CH_4: TDCS

To complete: TDCS of the CH_4 $1t_2$ and $2a_1$ orbitals with the same code, compared with Lahmam-Bennani *et al.* [11] (protocol objectives O2, O4). The code already contains the CH_4 target, so only the kinematics and the orbital switch need setting.

4.7 Post-collision interaction

To complete: Protocol objective O3: results with and without C_{se} (set $C_{se} = 1$ and its Gamow factor to 1) for the NH_3 orbitals at $E_e = 74$ eV and for CH_4. Discuss the shift of the binary peak and the change of the recoil-to-binary ratio.

4.8 Variable versus constant charge

To complete: Expected result 1: the same runs with $Z(r) \equiv 1$ for the outgoing electrons and $Z \equiv 0$ for the incident one (3C/BBK limit), to isolate the contribution of the variable charge.

5 Discussion: origin of the recoil deficit and domain of validity

5.1 The ejected-electron wave

The recoil peak comes from the ejected electron being back-scattered by the residual ion. Three results show that the ejected wave of Eq. (4) does not contain enough of it.

(i) Phase shifts. We solved the radial equation $u'' + [k^2 - l(l+1)/r^2 + 2Z_{ion}(r)/r]\, u = 0$ exactly with the same $Z_{ion}(r)$ at $E_e = 74$ eV. The short-range phase shifts beyond the $Z = 1$ Coulomb wave are large (Table 3). Inside the molecule, the partial waves of the 3CWZ wave keep the $Z = 1$ Coulomb phases, and their radial nodes are displaced. The local $\alpha(r)$ in Eq. (4) therefore does not produce a solution of the equation it approximates.

(ii) Back-scattering. The elastic cross section of the electron on NH_3^+ obtained with these phase shifts exceeds the Rutherford ($Z = 1$) value by factors 29–137 between 120° and 180°. This is the process the recoil peak requires.

(iii) Not a structure effect. In a plane-wave picture the binary/recoil ratio would be $\rho(q_{\mathrm{binary}})/\rho(q_{\mathrm{recoil}})$, with $q \approx 1.5$ and 3.2 a.u. here. This gives 38, 59 and 155 for $3a_1$, 1e and $2a_1$, against 1–2.5 measured. No realistic orbital can account for the observed recoil.

These findings agree with the failure of the two corrections tested in Section 4. The short-range potential acts mainly through the spurious non-orthogonality. Orthogonalization removes the spurious part but adds no back-scattering.

Table 3: Short-range phase shifts and elastic back-scattering for $e^- + NH_3^+$ at 74 eV.

l	0	1	2	3	4	5
δ_l (rad)	2.60	1.34	0.48	0.26	0.17	0.09
angle	30°	60°	90°	120°	150°	180°
$\sigma_{el}/\sigma_{\mathrm{Ruth}}(Z=1)$	2.5	4.8	4.9	29	91	137

5.2 Domain of validity

Within the present implementation, the molecular 3CWZ model is reliable for:

- the position and shape of the binary peak in asymmetric kinematics with small momentum transfer;
- the momentum profiles of EMS, at the level of the PWBA.

It is not reliable for:

- the recoil region and large ejection angles, because of the missing short-range distortion of the ejected electron;
- fast outgoing electrons or large momentum transfer, without grids adapted to K.

To complete: Revise this section after the CH_4, PCI and constant-charge results.

6 Conclusions and perspectives

Conclusions.

- We extended the 3CWZ model to molecular targets described by Moccia one-centre orbitals. The orientation average is analytic, and the effective charges are derived from the Hartree potential with a self-interaction correction.
- We implemented the model in a validated parallel code and applied it to the three valence orbitals of NH_3.
- The binary peak is well described.
- The recoil peak is underestimated by a factor 5–10. We traced this to the ejected-electron wave, not to the target description or to the numerics.

Perspectives.

- Replace the ejected 3CWZ wave by the exact distorted wave in the same Hartree potential (partial waves up to $l \approx 10$, matched to the $Z = 1$ Coulomb wave). This keeps the rest of the 3CWZ structure, in the spirit of the 1DW calculations for H_2O [14] and the distorted-wave methods [15].
- Use grids adapted to the momentum transfer, then compute DDCS, SDCS and total cross sections (comparison with Ref. [13]).
- Include the exchange amplitude and compare with the low-energy data of Nixon *et al.* [12].
- Extend the model to other molecules of interest (H_2O, biomolecular fragments).

To complete: Add the conclusions on CH_4 and on the PCI study.

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